

From Lucas Thresholds to Divisor Records and CRT Structure:

Exact Ordering Laws for the $mr \pm 1$ Factor-Witness Family

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Abstract

This paper records the continuation of an experimental number-theory program that originally began with an arithmetic-sequence construction, was reduced to binomial coefficients and Lucas-theoretic threshold predicates, and subsequently moved toward a different representation of the same semiprime factor geometry. The new development replaces the earlier binomial-product viewpoint by a family of congruences of the form

$$k^2 t^2 \equiv 1 \pmod{r},$$

which, for a prime factor r , reduces to

$$kt = mr \pm 1.$$

This representation yields an exact divisor optimization problem: for fixed multiplier m and search bound K , the best candidate is determined by the largest divisor $k \leq K$ of one of the integers $mr \pm 1$. A normalized score k/m then emerges naturally. Extensive computational experiments show exact agreement between this normalized ordering and the true ordering over the tested ranges.

The principal algebraic result is an exact comparison formula. For two witnesses

$$t_i = \frac{m_i r + e_i}{k_i}, \quad e_i \in \{\pm 1\},$$

with normalized ordering $k_1/m_1 > k_2/m_2$, define

$$\Delta = m_2 k_1 - m_1 k_2 > 0, \quad C = e_1 k_2 - e_2 k_1.$$

Then

$$t_1 < t_2 \iff \Delta r > C,$$

and the exact threshold is

$$r_{\min} = \left\lfloor \frac{C}{\Delta} \right\rfloor + 1.$$

Because $C \leq k_1 + k_2$ and $\Delta \geq 1$, one obtains the universal sufficient condition

$$r > k_1 + k_2 \implies t_1 < t_2.$$

Experiments further show that the bound is sharp: all observed equality cases satisfy $\Delta = 1$ and $(e_1, e_2) = (+1, -1)$, giving $r_{\min} = k_1 + k_2 + 1$. In this sharp family the witness divisibility conditions imply the congruences

$$r \equiv k_2 \pmod{k_1}, \quad r \equiv k_1 \pmod{k_2},$$

and hence a CRT description modulo $\text{lcm}(k_1, k_2)$. The computational evidence presented here strongly supports a divisor-record interpretation of the winner process, while leaving the full characterization of all winner transitions as an open problem.

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1 Introduction and Relation to the Previous Paper

The preceding paper developed an arithmetic-sequence factor-probing construction whose structured coefficient pattern reduced to binomial coefficients and, in a quotient-one regime, to an exact Lucas threshold predicate. The resulting procedure supported binary localization inside safe intervals and was then optimized computationally through cumulative products, paired products, and Montgomery multiplication. The principal unresolved difficulty was the cost of evaluating long numerator intervals without losing the factor-localization structure.

The present paper records a substantial change of direction. Rather than continuing to optimize the Lucas/binomial probe, the subsequent experiments returned to the algebraic geometry of the semiprime

$$N = pq, \quad p \leq q,$$

and sought lower-complexity identities whose factor divisibility could be studied directly.

The resulting chain is

$$\begin{aligned} \text{arithmetic-sequence construction} &\longrightarrow \text{quadratic factor identity} \longrightarrow k^2 t^2 - 1 \longrightarrow kt = mr \pm 1 \\ &\longrightarrow \text{divisor records} \longrightarrow \frac{k}{m} \text{ ordering} \longrightarrow \text{exact threshold} \\ &\longrightarrow \text{CRT structure.} \end{aligned}$$

The distinction between exact algebra and computational evidence is maintained throughout. Whenever a statement follows immediately from the displayed identities, it is treated as a mathematical result. Statements about the observed winner process, sharp cases, and frequency distributions are explicitly identified as experimental observations.

2 The Semiprime Coordinate and the Quadratic Reduction

Let

$$N = pq, \quad s = \lfloor \sqrt{N} \rfloor, \quad D = N - s^2.$$

A continuation of the arithmetic-sequence experiments produced the quadratic

$$H(x) = 4x^2 - (s + 4)x + 3D - 3s.$$

The hidden factor coordinate for a prime factor p is

$$x_p = s + 1 - p.$$

Substitution gives the exact identity

$$H(s + 1 - p) = p(4p + 3q - 7s - 4). \tag{1}$$

Thus the polynomial value at the hidden coordinate is divisible by the corresponding factor.

Introducing

$$k = s + 1 - x$$

gives

$$Q(k) = 4k^2 - (7s + 4)k + 3N. \tag{2}$$

For a factor $r \mid N$,

$$Q(k) \equiv k(4k - (7s + 4)) \pmod{r}. \tag{3}$$

This factorized modular form suggested searching for constructions in which divisibility reduced to a quadratic congruence with a particularly simple solution set.

3 A Second-Order Construction

A second construction considered expressions of the form

$$E(z) = (s - z)^2 R \left(z, \frac{z^2 + D}{s - z} \right). \quad (4)$$

For the choice

$$R = u + z + 1,$$

one obtains

$$E(z) = (s - z)(D + s + z(s - 1)). \quad (5)$$

Writing $t = s - z$ yields

$$E(s - t) = t(N - t(s - 1)), \quad (6)$$

and consequently

$$\gcd(N, E(s - t)) = \gcd(N, t^2(s - 1)). \quad (7)$$

Although this construction is not the final object studied in the present paper, it was useful in demonstrating that the original arithmetic-sequence setting contained much simpler modular divisibility structures than the original binomial formulation suggested.

4 The $k^2t^2 - 1$ Family

The main new direction begins with

$$k^2t^2 - 1. \quad (8)$$

Let r be an odd prime. Then

$$r \mid k^2t^2 - 1 \quad (9)$$

if and only if

$$(kt)^2 \equiv 1 \pmod{r}. \quad (10)$$

Since the only square roots of 1 modulo an odd prime are ± 1 ,

$$kt \equiv \pm 1 \pmod{r}. \quad (11)$$

Therefore there is an integer $m \geq 1$ and a sign $e \in \{+1, -1\}$ such that

$$kt = mr + e. \quad (12)$$

Hence the witness variable is

$$\boxed{t = \frac{mr + e}{k}}. \quad (13)$$

This representation isolates three discrete parameters:

- the multiplier m ;
- the divisor-like parameter k ;
- the sign $e \in \{\pm 1\}$.

The factor condition is now converted into a divisibility problem involving the much simpler integers $mr \pm 1$.

5 Modular Inverse Interpretation

When $\gcd(k, r) = 1$, the congruence

$$kt \equiv \pm 1 \pmod{r}$$

has the inverse representation

$$t \equiv \pm k^{-1} \pmod{r}.$$

If

$$u = k^{-1} \pmod{r},$$

then the two least positive residue representatives are u and $r - u$. Thus the relevant inverse signal can be written as

$$\rho_r(k) = \min(u, r - u). \quad (14)$$

Experiments in the transition from the modular-inverse description to the multiplier description found exact reconstruction agreement over the tested cases. This motivated replacing inverse calculations by the more concrete divisor formulation developed next.

6 Divisor-Record Formulation

Fix r , a multiplier m , and a bound K . Define

$$d_m(K) = \max\{k \leq K : k \mid mr + 1 \text{ or } k \mid mr - 1\}. \quad (15)$$

The corresponding best candidate for the multiplier m is then of the form

$$t_m(K) = \frac{mr + e}{d_m(K)} \quad (16)$$

for an appropriate sign e giving the selected divisor.

Thus the optimization problem is no longer an abstract modular inverse search. It is a record process over divisors of the two integers

$$mr - 1, \quad mr + 1.$$

Computational Experiment 399 independently reconstructed the exact winner through this divisor formulation and found no reconstruction failures or verification failures over its tested range. Experiment 403 then tracked changes in the winning multiplier as K increased. In the tested domain, every winner change coincided with the appearance of a new divisor-record event; no winner change was observed without such an event.

This motivates the terminology

Definition 1. *The divisor-record process is the evolution, as K increases, of the largest admissible divisor $k \leq K$ drawn from the numbers $mr \pm 1$, together with the multiplier m producing the best resulting witness.*

The present paper does not claim a theorem characterizing the complete record process; that remains an open problem.

7 The Normalized Divisor Score

The witness equation

$$t = \frac{mr + e}{k}$$

shows that for large r , the dominant scale of t is approximately

$$t \approx r \frac{m}{k}.$$

Thus minimizing t suggests maximizing

$$\boxed{S = \frac{k}{m}.} \tag{17}$$

For two witnesses

$$A = (m_1, k_1, e_1), \quad B = (m_2, k_2, e_2),$$

the normalized ordering is

$$\frac{k_1}{m_1} > \frac{k_2}{m_2}. \tag{18}$$

Experiments 404 and 405 found exact agreement between the normalized winner and the exact winner throughout their respective test domains. Experiment 406 extended the comparison to all primes $r \leq 5000$ with $K \leq 200$ and $m \leq 32$; no exact-ordering mismatch was observed. Experiment 407 subsequently demonstrated that mismatches do occur when the small- r boundary region is included more aggressively, leading to the exact correction analysis of the next section.

The normalized score is therefore not universally exact at finite r , but its failure has a completely explicit algebraic description.

8 Exact Pairwise Ordering

Consider two valid witnesses

$$t_1 = \frac{m_1 r + e_1}{k_1}, \quad t_2 = \frac{m_2 r + e_2}{k_2}, \quad e_1, e_2 \in \{\pm 1\}. \tag{19}$$

Assume that the normalized score ranks A first:

$$\frac{k_1}{m_1} > \frac{k_2}{m_2}. \tag{20}$$

Equivalently,

$$\Delta := m_2 k_1 - m_1 k_2 > 0. \tag{21}$$

Define also

$$C := e_1 k_2 - e_2 k_1. \tag{22}$$

Theorem 1 (Exact ordering criterion). *Under the normalized-first condition $\Delta > 0$,*

$$\boxed{t_1 < t_2 \iff \Delta r > C.} \tag{23}$$

Consequently the smallest integer r for which the normalized ordering becomes exact is

$$\boxed{r_{\min} = \left\lfloor \frac{C}{\Delta} \right\rfloor + 1.} \tag{24}$$

Proof. Cross-multiplication gives

$$\frac{m_1 r + e_1}{k_1} < \frac{m_2 r + e_2}{k_2}$$

if and only if

$$(m_1 r + e_1)k_2 < (m_2 r + e_2)k_1.$$

Rearranging,

$$(m_2 k_1 - m_1 k_2)r > e_1 k_2 - e_2 k_1,$$

which is exactly

$$\Delta r > C.$$

The threshold formula follows immediately. $\square$

This theorem explains the computational observations from Experiments 401–408: the normalized score controls the leading coefficient of r , while the signs e_i contribute only a bounded additive correction.

9 A Universal Ordering Bound

Because $e_i \in \{\pm 1\}$,

$$C = e_1 k_2 - e_2 k_1 \leq k_1 + k_2. \quad (25)$$

Since $\Delta \geq 1$ is an integer,

$$\frac{C}{\Delta} \leq k_1 + k_2. \quad (26)$$

Therefore the exact threshold satisfies

$$r_{\min} \leq k_1 + k_2 + 1. \quad (27)$$

More importantly, for integer r ,

Corollary 1. *If*

$$\frac{k_1}{m_1} > \frac{k_2}{m_2}$$

and

$$r > k_1 + k_2,$$

then

$$t_1 < t_2.$$

The proof is immediate: $r > k_1 + k_2 \geq C \geq C/\Delta$, hence $\Delta r > C$.

10 Experimental Verification of the Universal Bound

Experiment 409 used

$$r \leq 5000, \quad K \leq 500, \quad m \leq 64.$$

The computation covered

$$669$$

primes and

$$674,352,000$$

pair comparisons. The principal results were

$$\begin{aligned}\text{GLOBAL_BOUND_FAILURES} &= 0, \\ \text{MISMATCHES_R_ABOVE_2K} &= 0.\end{aligned}$$

The same experiment initially tested the stronger-looking threshold inequality $r_{\min} \leq k_1 + k_2$, which failed in a finite boundary population. This was not a failure of the ordering theorem. The correct general threshold is

$$r_{\min} \leq k_1 + k_2 + 1.$$

11 Sharpness of the Bound

Experiment 410 tested the corrected upper bound over the same prime, multiplier, and divisor ranges. It recorded

$$503,351,168$$

normalized-first pair tests and found

$$\text{THRESHOLD_BOUND_PLUS_ONE_FAILURES} = 0.$$

There were

$$86,627$$

cases in which equality held:

$$r_{\min} = k_1 + k_2 + 1. \tag{28}$$

Every such case satisfied

$$\Delta = 1, \tag{29}$$

$$C = k_1 + k_2. \tag{30}$$

The computational counters reported zero violations of either condition.

Moreover, every observed sharp case had the same sign pattern:

$$\boxed{e_1 = +1, \quad e_2 = -1.} \tag{31}$$

No sharp cases with the sign patterns $(+, +)$, $(-, +)$, or $(-, -)$ were observed in the tested range.

Thus the observed sharp family has the form

$$\boxed{m_2 k_1 - m_1 k_2 = 1, \quad e_1 = +1, \quad e_2 = -1.} \tag{32}$$

For this family,

$$C = k_2 + k_1 \tag{33}$$

and hence

$$r_{\min} = k_1 + k_2 + 1. \tag{34}$$

12 A Concrete Sharp Example

Experiment 410 produced the first sharp example

$$r = 7,$$

with

$$(m_1, k_1, e_1) = (2, 3, +1), \quad (m_2, k_2, e_2) = (3, 4, -1).$$

Indeed,

$$m_2 k_1 - m_1 k_2 = 3 \cdot 3 - 2 \cdot 4 = 1, \quad (35)$$

and

$$C = 3 + 4 = 7. \quad (36)$$

Therefore

$$r_{\min} = 8.$$

At $r = 7$ the two exact witnesses coincide:

$$\begin{aligned} t_1 &= \frac{2 \cdot 7 + 1}{3} = 5, \\ t_2 &= \frac{3 \cdot 7 - 1}{4} = 5. \end{aligned}$$

But their normalized scores satisfy

$$\frac{3}{2} > \frac{4}{3}.$$

At $r = 8$,

$$\begin{aligned} t_1 &= \frac{17}{3}, \\ t_2 &= \frac{23}{4}, \end{aligned}$$

so $t_1 < t_2$.

This example illustrates that the normalized order can fail exactly at the finite correction boundary, while becoming correct immediately beyond it.

13 The Exact Spacing Law in the Sharp Family

For the sharp family

$$\Delta = 1, \quad e_1 = +1, \quad e_2 = -1,$$

the difference between the exact witnesses simplifies to

$$\boxed{t_2 - t_1 = \frac{r - (k_1 + k_2)}{k_1 k_2}}. \quad (37)$$

Indeed,

$$t_2 - t_1 = \frac{(m_2 r - 1)k_1 - (m_1 r + 1)k_2}{k_1 k_2},$$

and using

$$m_2 k_1 - m_1 k_2 = 1$$

gives the stated identity.

Therefore the formal crossing point is exactly

$$r = k_1 + k_2.$$

The actual witness construction additionally requires $k_1 \mid m_1 r + 1$ and $k_2 \mid m_2 r - 1$; consequently the formal threshold need not itself be an admissible prime witness.

14 CRT Structure of the Sharp Family

Suppose

$$m_2 k_1 - m_1 k_2 = 1, \quad (38)$$

and suppose the sharp witnesses are valid with signs $(+1, -1)$. Then

$$k_1 \mid m_1 r + 1, \quad k_2 \mid m_2 r - 1. \quad (39)$$

Because $\Delta = 1$,

$$\gcd(m_1, k_1) = \gcd(m_2, k_2) = 1.$$

Reducing the determinant relation modulo k_1 gives

$$-m_1 k_2 \equiv 1 \pmod{k_1},$$

so

$$m_1 k_2 \equiv -1 \pmod{k_1}.$$

The witness congruence gives

$$m_1 r \equiv -1 \pmod{k_1}.$$

Since m_1 is invertible modulo k_1 ,

$$\boxed{r \equiv k_2 \pmod{k_1}.} \quad (40)$$

Similarly, reducing the determinant relation modulo k_2 gives

$$m_2 k_1 \equiv 1 \pmod{k_2},$$

while the second witness condition gives

$$m_2 r \equiv 1 \pmod{k_2}.$$

Hence

$$\boxed{r \equiv k_1 \pmod{k_2}.} \quad (41)$$

Therefore every observed sharp witness satisfies the pair of congruences

$$\boxed{r \equiv k_2 \pmod{k_1}, \quad r \equiv k_1 \pmod{k_2}.} \quad (42)$$

When these congruences are compatible, the Chinese remainder theorem gives a unique residue class modulo

$$L = \text{lcm}(k_1, k_2).$$

Equivalently one may write the condition as

$$r \equiv k_1 + k_2 \pmod{L}, \quad (43)$$

with the usual interpretation that this compact form represents the simultaneous component congruences.

15 Experimental Verification of the CRT Structure

Experiment 411 enumerated determinant-one pairs satisfying

$$m_2k_1 - m_1k_2 = 1$$

within

$$m_i \leq 64, \quad k_i \leq 500,$$

and tested their relation to prime residues up to 5000.

The full unconstrained congruence count was intentionally larger than the reverse-construction population; arbitrary determinant-one pairs need not correspond to a prime r satisfying the required residue conditions. The meaningful conditional tests were:

$$\begin{aligned} \text{REVERSE_TESTS} &= 130,560, \\ \text{REVERSE_FAILURES} &= 0, \\ \text{WITNESS_FAILURES} &= 0, \\ \text{T_DIFFERENCE_FAILURES} &= 0. \end{aligned}$$

Independently, the actual sharp cases observed in the previous experiment were checked against the CRT conditions. The result was

$$\begin{aligned} \text{OBSERVED_SHARP_CASES} &= 86,627, \\ \text{OBSERVED_SHARP_CONGRUENCE_FAILURES} &= 0, \\ \text{OBSERVED_SHARP_SIGN_PM_FAILURES} &= 0. \end{aligned}$$

These results strongly support the CRT description of the observed sharp family. They do not, by themselves, prove that every determinant-one pair is realized by a prime witness, nor do they characterize the complete set of winner transitions.

16 Summary of Experiments

Experiment	Main purpose	Principal result
393	Modular-inverse verification	The inverse-based and witness-based formulations agreed except for explicitly understood shared-factor edge cases; valid roots verified.
397–399	Multiplier/divisor reconstruction	The forms $kt = mr \pm 1$, modular inverses, and largest-divisor reconstruction produced the same winners on the tested instances.
400	$m = 1$ to $m = 2$ transitions	Multiplier $m = 2$ begins to overtake $m = 1$ at finite K , demonstrating that the $m = 1$ family is not universally dominant.
401	Ratio transition criterion	A normalized divisor-ratio condition predicted observed $m = 1 \rightarrow 2$ transitions without observed failures.
402	Winner progression	Winners progressed through increasing multiplier families in the tested ranges, with no backward winner transitions observed.
403	Record-event structure	Every observed winner change occurred at a new divisor-record event.

Experiment	Main purpose	Principal result
404–405	Normalized score	The score k/m matched exact winners over large tested populations.
406–408	Boundary analysis	Small- r mismatches were found and then exactly described by the correction formula $\Delta r > C$.
409	Universal ordering bound	No failures for $r > k_1 + k_2$ over 674,352,000 pair comparisons.
410	Sharpness	No failures of $r_{\min} \leq k_1 + k_2 + 1$ over 503,351,168 normalized-first comparisons; all 86,627 equality cases had $\Delta = 1$ and signs $(+, -)$.
411	CRT structure	All observed sharp cases satisfied the predicted congruences; the conditional reverse construction had zero failures.

17 What Is Proven and What Remains Empirical

It is useful to separate the current state into three levels.

17.1 Exact algebraic results

The following statements follow directly from the definitions:

1. $r \mid k^2 t^2 - 1$ implies $kt = mr \pm 1$ for prime r .
2. Every witness in the multiplier family has the form

$$t = (mr + e)/k.$$

3. For normalized-first pairs,

$$t_1 < t_2 \iff \Delta r > C.$$

4. The exact threshold is

$$r_{\min} = \lfloor C/\Delta \rfloor + 1.$$

5. The universal sufficient condition

$$r > k_1 + k_2$$

guarantees exact agreement with the normalized order.

6. In the determinant-one $(+, -)$ family,

$$t_2 - t_1 = \frac{r - (k_1 + k_2)}{k_1 k_2}.$$

7. The sharp witness divisibility conditions imply the CRT congruences stated above.

17.2 Computationally supported observations

The following have been repeatedly observed but are not claimed here as theorems solely from the experiments:

1. the normalized score k/m reproduces the exact winner over the tested large domains;
2. winner changes occur at new divisor-record events;
3. the sharp boundary is always realized through $\Delta = 1$ and the $(+, -)$ sign pattern within the tested domain;
4. the observed sharp cases satisfy the corresponding CRT residue classes;
5. the entire winner process may admit a complete divisor-record characterization.

18 Open Problems

The present results suggest several concrete mathematical questions.

18.1 Complete characterization of winner transitions

For fixed prime r , bound K , and multipliers m , can the winning candidate be characterized solely through divisor records of $mr \pm 1$? Experiment 403 suggests this strongly, but a proof for the complete process is not yet available.

18.2 Classification of sharp determinant-one families

The observed sharp cases satisfy

$$m_2 k_1 - m_1 k_2 = 1, \quad e_1 = +1, \quad e_2 = -1.$$

A natural next step is to classify all integer quadruples satisfying the determinant-one relation together with the witness divisibility conditions, including their CRT compatibility.

18.3 Prime realization of CRT classes

Given a determinant-one pair (m_1, m_2, k_1, k_2) , the sharp family asks for primes in the CRT residue class

$$r \equiv k_2 \pmod{k_1}, \quad r \equiv k_1 \pmod{k_2}.$$

The arithmetic conditions on the residue class, its modulus, and its prime realizability should be studied systematically.

18.4 From pairwise ordering to factor localization

The ultimate question remains algorithmic: can the divisor-record/normalized-score structure be exploited to localize an unknown prime factor of N without an essentially linear scan over a large witness space?

The current work has simplified the local comparison problem dramatically, but it has not yet produced a proven general-purpose factoring algorithm.

19 Conclusion

The post-Lucas continuation has transformed the original factor-probing problem into a substantially simpler arithmetic comparison problem. The central witness family

$$kt = mr \pm 1$$

turns factor detection into a divisor problem over the neighboring integers $mr - 1$ and $mr + 1$. The best candidate for a multiplier is controlled by the normalized divisor score k/m , while the exact finite- r discrepancy is completely described by the pair

$$\Delta = m_2 k_1 - m_1 k_2, \quad C = e_1 k_2 - e_2 k_1.$$

The exact comparison law

$$t_1 < t_2 \iff \Delta r > C$$

explains the previously observed normalized behavior and yields the clean global guarantee

$$r > k_1 + k_2 \implies \text{normalized order} = \text{exact order}.$$

The corresponding threshold satisfies

$$r_{\min} \leq k_1 + k_2 + 1,$$

and the computational study found 86,627 equality cases, all with determinant one and sign pattern $(+, -)$. Those sharp cases further collapse to a pair of modular congruences and hence to a CRT description.

The research has therefore moved from a Lucas-theoretic threshold problem toward a new arithmetic object: a divisor-record process constrained by normalized ratios and determinant-one transitions. The remaining challenge is to determine whether this local structure can be lifted into a global, factor-blind localization theorem.